

Preliminary Propulsion Report

EuRoC Engineering Team

March 24, 2025

Summary

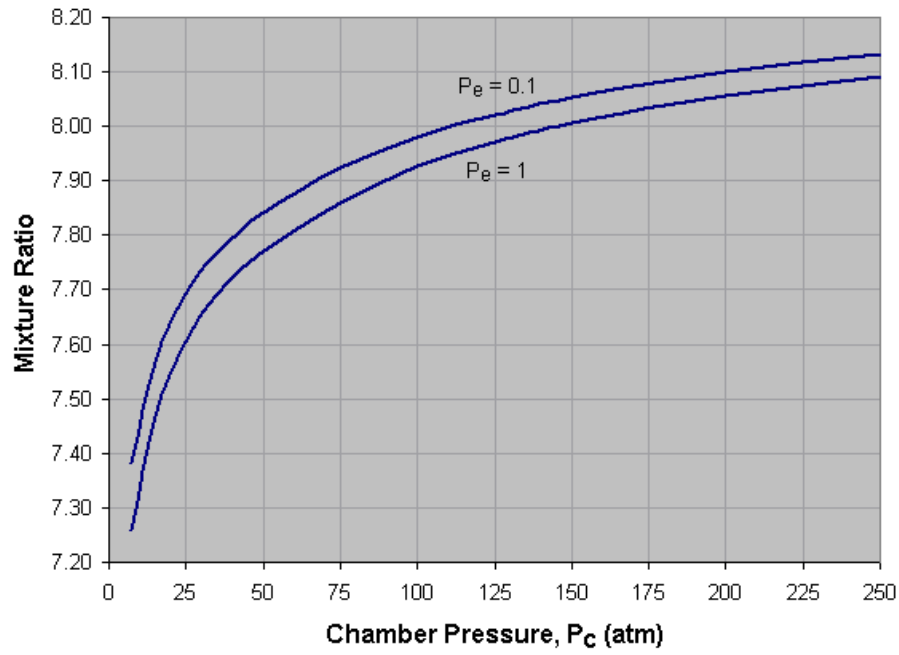
This technical report presents our current findings as of current date, as well as lays out the basis of further research and development on behalf of our team. Data was collected using the trial version of RPA (Rocket Propulsion Analysis), NASA's CEARUN, and a collection of publicly available online resources, primarily MIT's Liquid Propulsion Resources. Simulations are run at a centimeter scale, as a proof of concept.

Initial Parameters

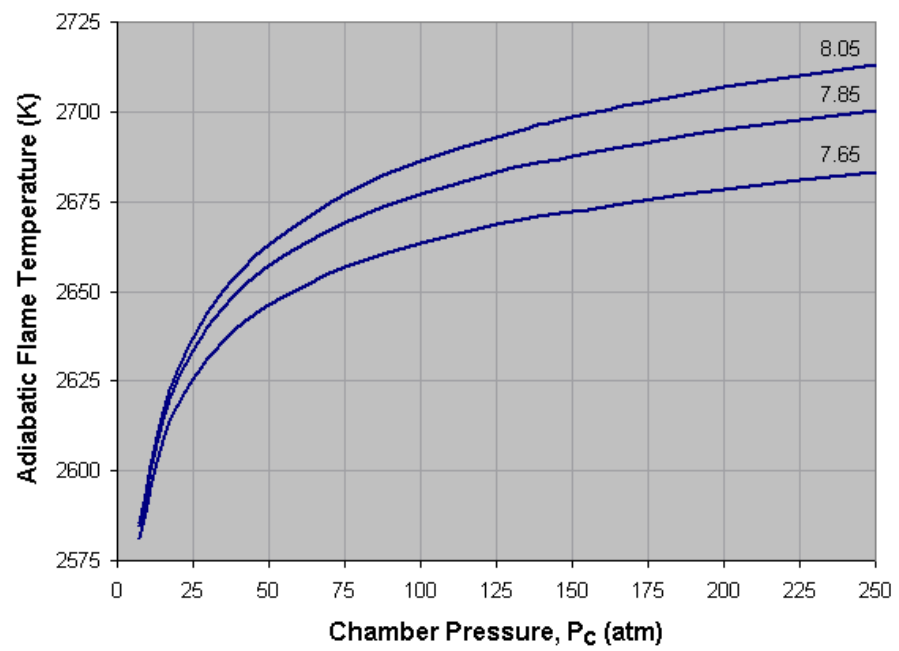
Upon consulting standard available literature, our team found that most common bipropellants are unideal for student teams such as ours, as Cryogenics are difficult to work with under nominal temperature standards, and Hypergolics are less than ideal due to increased toxicity and volatility. Thus, upon consulting estimated build costs and availability, and factoring in complexity of manufacture given our status as a SRAD propulsion team, we have settled upon a mixture of High-Test Peroxide (HTP) 85% as our oxidizer and RP-1 Kerosene as our propellant. An alternative most often taken by American teams is a mixture of Ethanol and Nitrous Oxide, however for our use case, given the lack of infrastructure within Romania to handle Nitrous and the lessened potential energy output of Ethanol, we opted for our current selection.

Mixture Properties

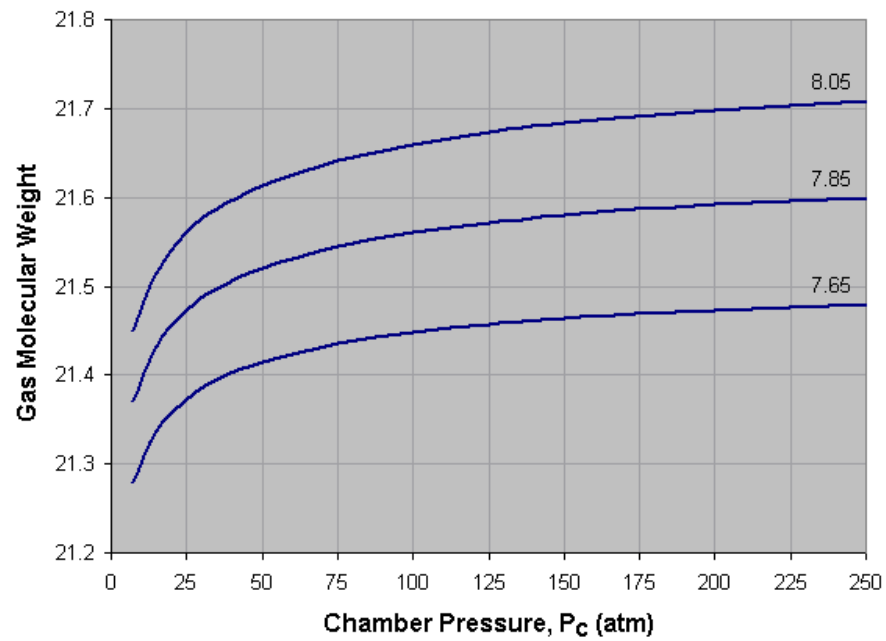
Assuming combustion chamber pressure to be 68 atm (1000 PSI) and nozzle exit pressure to be 1 atm, we can find our Oxidizer to Fuel (O/F) ratio to be 7.84, our specific impulse to be 258s at sea level, and our density impulse to be 324 kg-s/l, as given by "Ignition!", by John Drury Clark. Graphs are indicated below, courtesy of Robert A. Braeunig:



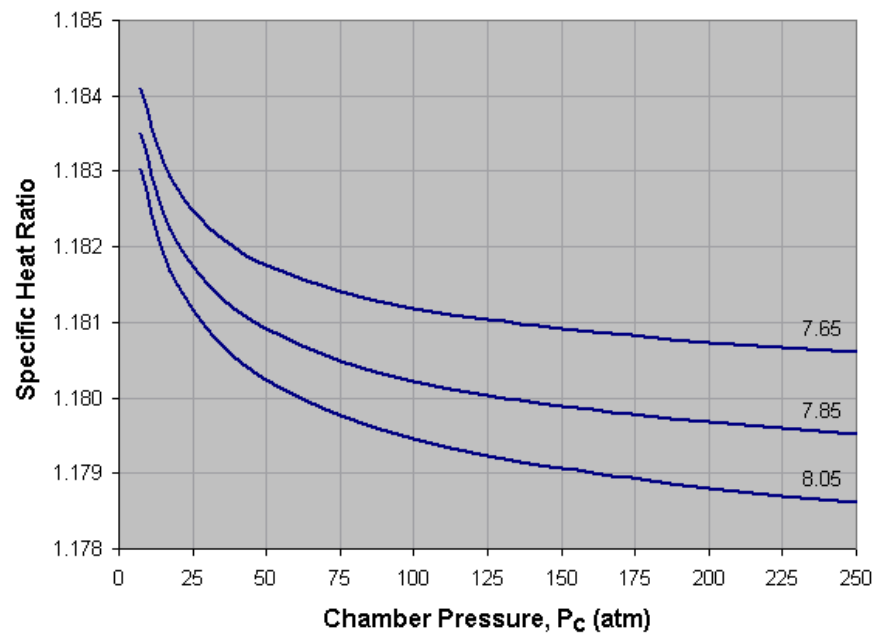
Optimum Mixture Ratio



Adiabatic Flame Temperature



Gas Molecular Weight



Specific Heat Ratio

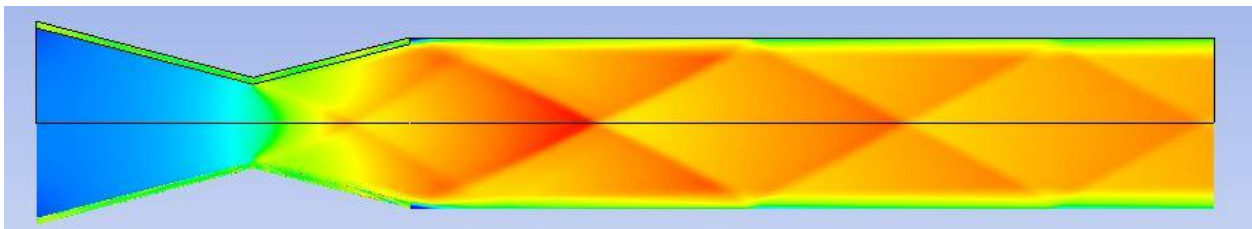
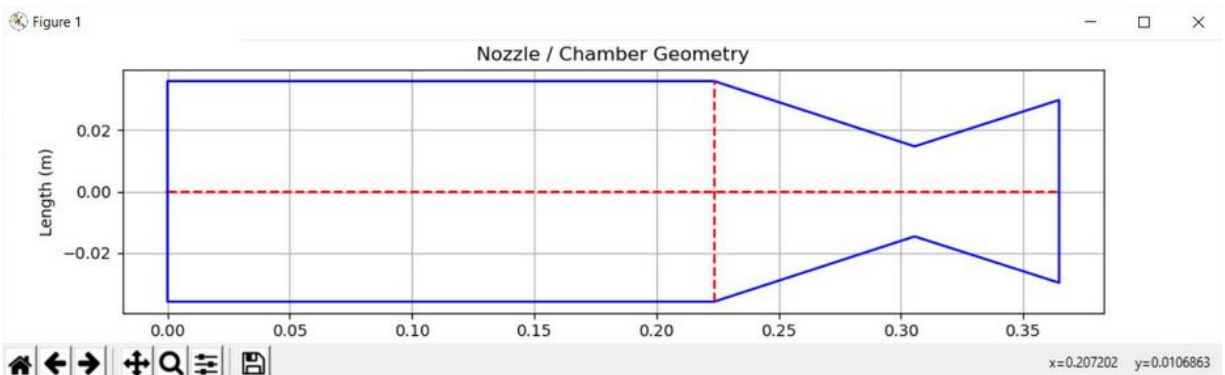
Simulation Results

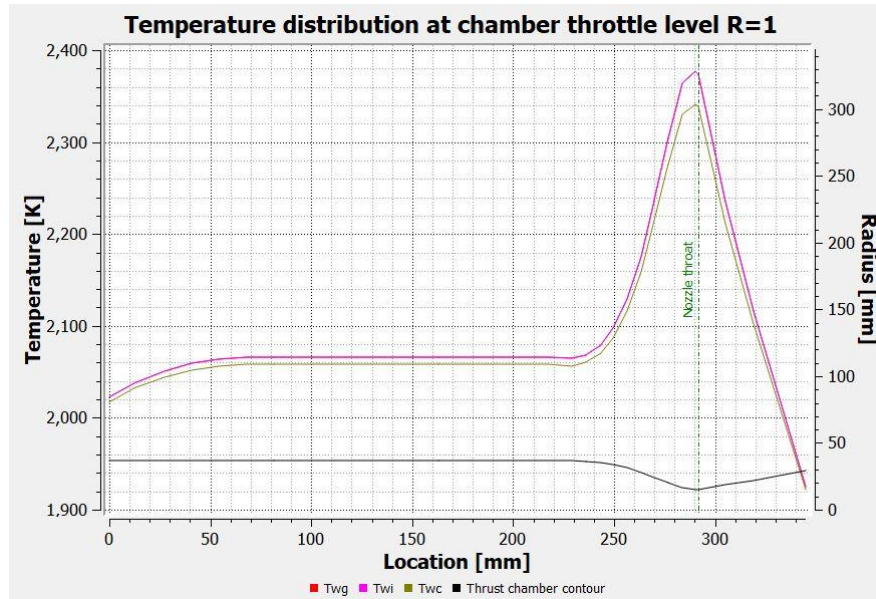
Assuming above stated results, CEARUN gives us the following:

O/F= 7.65000

	CHAMBER	THROAT	EXIT	PERFORMANCE PARAMETERS		
Pinf/P	1.0000	1.7249	20.414		THROAT	EXIT
P, BAR	20.684	11.992	1.0132	Ae/At	1.0000	4.0339
T, K	2871.56	2728.96	2105.20	CSTAR, M/SEC	1635.7	1635.7
RHO, KG/CU M	1.9280	0 1.1868	0 1.3378	CF	0.6514	1.4326
H, KJ/KG	-5086.80	-5654.50	-7832.21	Ivac, M/SEC	2013.8	2666.5
U, KJ/KG	-6159.61	-6664.95	-8589.61	Isp, M/SEC	1065.6	2343.3
G, KJ/KG	-39989.9	-38824.3	-33420.5			
S, KJ/(KG)(K)	12.1548	12.1548	12.1548			
M, (1/n)	22.255	22.455	23.110			
(dLV/dLP)t	-1.01562	-1.01265	-1.00229			
(dLV/dLT)p	1.3523	1.3011	1.0714			
Cp, KJ/(KG)(K)	5.4481	5.1084	3.1237			
GAMMA _s	1.1233	1.1237	1.1493			
SON VEL, M/SEC	1097.8	1065.6	933.0			
MACH NUMBER	0.000	1.000	2.512			

Similarly, a microscale motor in RPA gave us a temperature reading of approximately 2400K, leading us to believe that a solid material choice for the near future would be GRCo_p 42 or 84.





Future Milestones

As of current date, one of the most important priorities for our team would be to, after obtaining funding, purchase a copy of Rocket Propulsion Analysis, as the current trial version is limiting in many ways. Another immediate future milestone would be to dedicate more research into properly applying and understanding the CEARUN software supplied by NASA, for more effective application of this tool, alongside a comprehensive evaluation of both Rocket Propulsion Elements by George P. Sutton and Modern Engineering for Design of Liquid-Propellant Rocket Engines by D. Huzel and D. Huang.

More milestones at a later date would include 3D modelling and fluid simulation within SolidWorks, contracting a CNC milling firm to fabricate our parts, and our first static fire tests.